

CSA1717-ARTIFICIAL INTELLIGENCE PRACTICAL LAB EXPERIMENTS

MEDICAL DIAGNOSIS

Aim:

To implement a Prolog-based medical diagnosis system using symptoms and logical rules.

Objective:

To infer possible diseases based on symptoms provided by the user.

Algorithm

1. Start.
2. Define symptoms.
3. **Define disease rules.**
4. **Enter the patient's symptoms.**
5. **Match symptoms with disease rules.**
6. **Display the possible diagnosis.**
7. Stop.

Flowchart START

| v

Enter Symptoms

| v

Match Disease Rules

| v

Required Symptoms Present?

/ \

Yes No

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Disease Check Derived Next Rule

| v

Display Diagnosis

| v

STOP

Prolog Program

% Medical Diagnosis System

symptom(ravi, fever). symptom(ravi, cough). symptom(ravi, body_pain).

symptom(priya, fever). symptom(priya, rash).

symptom(arun, cough). symptom(arun, breathlessness).

diagnosis(Patient, flu) :-symptom(Patient, fever), symptom(Patient, cough).

diagnosis(Patient, measles) :-symptom(Patient, fever), symptom(Patient, rash).

diagnosis(Patient, pneumonia) :-symptom(Patient, cough), symptom(Patient, breathlessness).

Queries

?- diagnosis(ravi, Disease).

Output:

Disease = flu.

?- diagnosis(priya, Disease). Output:

Disease = measles.

?- diagnosis(arun, Disease). Output:

Result:

Thus, the medical diagnosis system successfully identifies possible diseases based on symptoms.